

PRACTICAL OF PC SET THREE
Higher National Diploma Exam-June-July 2020 Session

Speciality: **SOFTWARE ENGINEERING (SM)**

Paper: **PRACTICE OF COMPUTER**

Nature of the paper: **Practical**

Coefficient : **3**

Duration : **5hrs**

SECTION A : PROGRAMMING. (50marks)

Students are required to have the following software installed in their computers:

- Code block, for C and C++ programing
- Visual Studio, for Event programming
- Netbeans or Eclipse or JBlue for Java
- Wamp or Xampp for database

I- C- programming

(15mks)

Consider the sequence $S_n = \sum_{i=1}^n (x^i) = x^1 + x^2 + x^3 + \dots + x^n$

- i. Write a C code that allows the user to input a real value $0 < x < 1$, calculate the value of $n=N$ such that $S_N \leq 1$. N and S_N will be displayed. (5mks)
- ii. An error message (ERROR: the value of x is too small) that permits the user to know that the value of x was too small which would permit the program to enter an infinite loop was entered would also be printed. This would occur if the program attempts to do more than 50 iterations. (5mks)
- iii. Define an array called sum and put all the values of S_i , $1 \leq i \leq N$. The program would print out the values of S_i both vertically and horizontally. (5mks)
- iv. Write another set of codes on the same program that deletes the k th value in the array where $k < N$ and display the terms of the table where $k < N$. (5mks)

II- Object oriented programming

(20mks)

Write a program that defines a shape class with a constructor that gives value to width and height. Then define two sub-classes triangle and rectangle, that calculate the area of the shape area (). In the main, define two variables a triangle and a rectangle and then call the area() function in this two variables. Clearly show how inheritance is exhibited by sufficiently commenting the program. Use either C++ or Java.

III- Event programming

1- Create in ASP.NET a page as below:

(15mks)

Login

Password

Country

:

:

: Macedonia ▼

:

- Create another page and rename it as example1.aspx (5mks)
- When the user clicks on Login button in (a), the page should redirect to example1.aspx. (10mks)

NB: The Reset button is used to clear the Login and Password text fields.

SECTION B: Database Development and Administration (MySQL). (20marks)

Sectors (Sec#, sec_name, city)

Employees (Emp#, Emp_name, Qualification, Salary, Sec#*)

Projects (Project#, Project_name, Budget)

Participation (Emp#*, Project#*, Function)

Following the relational schema provided above where the attributes underlined are primary keys and those with the symbol (*) denotes foreign keys, consider developing a relational database using MySQL